

STAGE 1 · DATA AND PROTOCOL

Four clinical ECG sources

PTB-XL | Georgia | Chapman | Ningbo

66,379 retained records, 13 harmonised labels, 10 s at 100 Hz



leave-one-source-out: train on three, test on the fourth, four rotations

STAGE 2 · MODELS AND CONTROLLED BUDGETS

10 encoder families × 2 size budgets

compact ≤ 32 KB INT8 weights | standard ≤ 128 KB

each family scaled to the largest width that fits

everything else pinned: loss, optimiser, epochs, seed



80 trained models

STAGE 3 · RESEARCH QUESTIONS

RQ1

External discrimination

external macro-AUPRC and AUROC per held-out source, FP32 and INT8, record-level bootstrap intervals

Record-level intervals $\neq$ between-source uncertainty

RQ2

Hardware characterisation

ST Edge AI $\rightarrow$ arm-none-eabi
 $\rightarrow$ flash over SWD;
measured Flash, peak SRAM and DWT latency; 20 of 20 ran

MACC $\neq$ measured latency

RQ3

Performance-resource trade-offs

observed trade-offs across families and budgets;
non-dominated configurations among those tested

Pareto frontier is descriptive, not architecture search